Zhuoqian (Zack) Yang

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Education

École Polytechnique Fédérale de Lausanne, IVRL — Lausanne, CH
PhD Candidate, supervised by *Prof. Mathieu Salzmann* and *Prof. Sabine Süsstrunk* — Sep 2023 – Present
Expected graduation: Fall 2027 | Available for internship from Spring 2027

Carnegie Mellon University, Robotics Institute — Pittsburgh, PA, US
M.S. in Computer Vision, advisor: *Dr. László Jeni* | GPA: 4.03/4.3 — Aug 2019 – Dec 2020
Courses: Machine Learning (A+), Computer Graphics (A), Computer Vision (A+), Computational Photography (A-), Math for Robotics (A)

Beihang University, School of Software Engineering — Beijing, CN
B.E. in Software Engineering | GPA: 88.1/100, Ranking: 6/149 — Sep 2015 – Jun 2019

Publications

Full list on Google Scholar (1,171 citations):
https://scholar.google.com/citations?user=WTDthdcAAAAJ

* equal contribution

[1] **Weight Space Representation Learning via Neural Field Adaptation** — **CVPR 2026**
Zhuoqian Yang, Mathieu Salzmann, Sabine Süsstrunk
Project Page: https://inrainbws.github.io/wsr/

[2] **Subtractive Modulative Network with Learnable Periodic Activations** — **ICASSP 2026 (Oral)**
Tiou Wang, Zhuoqian Yang, Markus Flierl, Mathieu Salzmann, Sabine Süsstrunk
Project Page: https://inrainbws.github.io/smn/

[3] **3DHumanGAN: 3D-Aware Human Image Generation with 3D Pose Mapping** — **ICCV 2023**
Zhuoqian Yang, Shikai Li, Wayne Wu, Bo Dai
Project Page: https://3dhumangan.github.io/

[4] **OrthoPlanes: A Novel Representation for Better 3D-Awareness of GANs** — **ICCV 2023**
Honglin He*, Zhuoqian Yang*, Shikai Li, Bo Dai, Wayne Wu
Project Page: https://orthoplanes.github.io/

[5] **TransMoMo: Invariance-Driven Unsupervised Video Motion Retargeting** — **CVPR 2020**
Zhuoqian Yang*, Wentao Zhu*, Wayne Wu*, Chen Qian, Qiang Zhou, Bolei Zhou, Chen Change Loy
Project Page: https://inrainbws.github.io/transmomo

[6] **MEAD: A Large-Scale Audio-Visual Dataset for Emotional Talking Face Generation** — **ECCV 2020**
KSY Wang*, Qianyi Wu*, Linsen Song*, Zhuoqian Yang, Wayne Wu, ..., Chen Change Loy
Project Page: https://wywu.github.io/projects/MEAD/MEAD.html

Research Experience

EPFL, Image and Visual Representation Lab (IVRL), PhD Candidate — Lausanne, CH
Topic: Weight Space Representation Learning
Supervisor: *Prof. Mathieu Salzmann* and *Prof. Sabine Süsstrunk* — Sep 2023 – Present

- Showed that i) Properly constrained weights exhibit semantic structure; ii) Structured weight space enables effective weight space generation. Achieves the **first successful weight space generation** of natural images at 128^2 (CVPR 2026).
- Ongoing i) Weight space generation as test-time adaptation for large language models. ii) Data-agnostic generative model via weight space neural field generation.
- Supervising: Subtractive Modulative Network (ICASSP 2026 Oral), an implicit neural representation architecture inspired by the subtractive synthesis pipeline of audio synthesizers;
  ExpandDiff, a diffusion model trained to restore dynamic range removed by synthetic clipping, for single-image high dynamic range (HDR) reconstruction.

Shanghai AI Laboratory, Research Engineer — Shanghai, CN

Topic: 3D-Aware Representation and Generation

Supervisor: *Dr. Bo Dai* and *Dr. Wayne Wu* — Mar 2021 – Jul 2023

- Proposed one of the first 3D-aware generative adversarial network 3DHumanGAN that synthesizes full-body human images; achieved pose-conditioned 3D-aware synthesis of full-body images with image quality comparable to that of 2D GANs.
- Proposed a novel hybrid implicit-explicit volumetric representation OrthoPlanes for neural rendering of 3D scenes; significantly improves quality and view-consistency of rendering on large view-angles.

CMU and Fujitsu Research of America, Master's Student — Pittsburgh, PA, US

Topic: Semantic Facial Image Manipulation using 2D/3D Modalities, *sponsored capstone project*

Supervisor: *Dr. László Jeni* and *Dr. Koichiro Niinuma* — May 2020 – Dec 2020

Presentation: https://drive.google.com/file/d/1UbTcLudyGlr4OiP10rQ53OjattXGkA5B

- Built a facial expression manipulation model to generate photorealistic images conditioned on Facial Action Coding System (FACS) action units.
- Designed a two-stage pipeline: (i) manipulate image geometry using 3D information of the face, (ii) synthesize facial-expression-induced textures such as wrinkles.
- Improved FACS expression accuracy by 19% and Fréchet Inception Distance (FID) by 12%.

SenseTime Research, Research Intern, *Motion Retargeting* (CVPR 2020, AAAI 2021) — Mar 2019 – Dec 2019

Beihang University, Research Student, *Visual Question Answering* (ICIP 2020) — Jul 2018 – Dec 2018

Yunnan Normal University, Research Student, *Non-Rigid Image Registration* (IEEE TIP) — Mar 2017 – Dec 2017

Software

ColorHead, free desktop app for color negative inversion (macOS, Windows) — Jun 2026

Website: https://colorhead.app

Turns scanned color negatives into accurate positives with a pipeline that physically simulates analog printing: statistically estimates the film's parameters from a whole roll, simulates the response curve of photographic paper, and offers enlarger-style CMY color grading as well as other analog-based adjustments. Supports camera scans and film scanners.

Skills

Research: generative models (diffusion models, autoregressive models, GANs), weight space learning, neural fields, Large Language Model (LLM) post-training, Low-Rank Adaptation (LoRA), test-time adaptation, 3D-aware generation, neural rendering, human motion, computer vision, computer graphics, signal processing

Engineering: Python, PyTorch, JAX, CUDA, Hugging Face, SLURM

Awards

- Competitive Compute Grant, ~CHF 40,000, Swiss National Supercomputing Centre (CSCS) — Jun 2026
- EDIC PhD Fellowship, CHF 54,000, EPFL — Mar 2023
- First Prize Scholarship for Academic Performance (Top 5%), Beihang University — Dec 2016

Academic Services

Reviewer: CVPR, ECCV, ICCV, ICLR, AAAI, ACM MM, ICASSP

Volunteer, ICCP 2024 — Lausanne, CH

Teaching Assistant, EPFL

Course *CS-341 Computer Graphics* — Jan 2024 – May 2024

Course *CS-233 Introduction to Machine Learning* — Jan 2026 – May 2026

Talks

Weight Space Representation Learning via Neural Field Adaptation — Feb 2026

- AI:ML Group Seminar, University of St. Gallen